The Resolution Calibration Principle: Certified Locality for Kernel and Gibbs Fields

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Abstract

Distance-weighted predictors are treated as local, although every stored point can influence every query. We introduce the Resolution Calibration Principle, a label-free certificate that bounds the aggregate influence of sources beyond a chosen radius. For additive Gaussian fields, one conservative measurement of the far-field mass A gives the closed-form bandwidth $\sigma^* = d/\sqrt{2\ln(A/\varepsilon)}$, and the same bound certifies decision stability, abstention and insensitivity to deleting distant data. Across vision, text and tabular fields, σ^* holds the budget where label-free bandwidths exceed it by orders of magnitude, approaching labelled-search accuracy and re-calibrating as memories densify. A Gibbs form extends the principle to normalised readouts: a prototypical network is certified within budget, and on evidence-grounded question answering the certificate identifies attention heads with certified outside-evidence mass and bounded, empirically small deletion sensitivity, while a far-mass penalty trains a reader toward that locality. Resolution is thereby calibrated from geometry, making globally supported influence certifiably local.

Many machine-learning systems answer a query by aggregating influences from stored points. Kernel predictors, prototype classifiers and attention mechanisms differ in implementation, but share a structural feature: their influences decay with distance or energy without becoming exactly zero. A readout intended to behave locally is therefore globally supported. As memories densify, individually weak distant sources can accumulate, alter decisions and obscure which data determined an output. Locality must be established for the aggregate field, not inferred from the decay of one influence. The near region can also be supplied externally, by a task's own evidence annotation, and the certificate then reports whether the field respects that declared locality.

Bandwidths and temperatures are usually selected to optimise predictive criteria such as validation accuracy, likelihood or density error [1, 2, 3, 4]. These procedures can identify a high-performing operating point, but they do not answer the locality question: how much aggregate influence remains outside the region the system is intended to use? Hard truncation guarantees locality by changing the readout. Missing is a calibration rule that preserves the smooth field while converting a chosen radius and interference budget into an operating resolution.

We provide such a rule. In one line: measure how much total influence the distant sources carry, then pick the widest resolution at which that influence still falls below a declared budget. Concretely, for an additive Gaussian kernel-vote field of bandwidth σ (a peak-normalised kernel, not an absolute density), the influence beyond radius d is bounded by $A\exp(-d^2/2\sigma^2)$, where A is the effective mass of the far sources. Inverting this envelope gives the certified frontier $\sigma^* = d/\sqrt{2\ln(A/\varepsilon)}$: the largest bandwidth certified to keep aggregate far-field interference below ε. The mass A is measured from the field, not fitted to labels, and is nondecreasing in σ, so measuring it once at a broader reference scale is conservative. This is the Resolution Calibration Principle.

The certificate has direct operational consequences. At $\sigma \le \sigma^*$, deleting all far sources moves the output by at most ε; if the near-field margin exceeds ε, those sources cannot change the decision; and low-margin queries can be withheld. A Rényi/Hill family tightens the effective-mass bound. For normalised Gibbs fields, a companion temperature certificate controls far probability mass directly, covering prototype softmaxes and attention without the amplitude restriction of the additive form. A label-free conformal wrapper then turns the average-mass construction into finite-sample marginal coverage for a future query.

We test these claims on frozen vision, text and tabular fields, and on trained transformer and prototype readouts. The additive frontier holds the interference budget where the label-free bandwidth rules tested here exceed it by orders of magnitude, while approaching labelled grid-search accuracy. When a memory densifies, re-measuring A tracks a per-stage re-tuned oracle without labels. The Gibbs experiments expose both the reach and the diagnostic value of the principle: a prototypical network can be certified near its operating temperature, whereas on HotpotQA a fine-tuned reader's attention is diagnosed as largely nonlocal relative to annotated evidence, with a certified minority of heads on which the bound limits how far deleting outside-evidence keys can move the readout.

1 Results

1.1 Calibrating aggregate locality

A kernel field is a collection of sources $z_i \in \mathbb{R}^n$ with fixed signed weights v_i and a common Gaussian kernel

$$K_\sigma(t) = \exp(-t/2\sigma^2), \qquad F_\sigma(y) = \sum_i v_i K_\sigma(\|y - z_i\|^2). \tag{1}$$

For a query y, radius d separates near sources from the far set $\mathcal{F} = \{i : \|y - z_i\| > d\}$. The quantity to control is the aggregate far tail $\mathrm{Tail}_\sigma(y, d) = \sum_{i\in\mathcal{F}} |v_i| K_\sigma(\|y - z_i\|^2)$. The raw number of far sources is unnecessarily pessimistic when only a few contribute. We instead use their participation ratio

$$N_{\mathrm{eff}}(\sigma) = \frac{(\sum_{i\in\mathcal{F}} \omega_i)^2}{\sum_{i\in\mathcal{F}} \omega_i^2}, \qquad \omega_i = K_\sigma(\|y - z_i\|^2), \tag{2}$$

and define $A = V_{\max} N_{\mathrm{eff}}$, with $|v_i| \le V_{\max}$. The participation ratio is computed on the far set: including near sources would make it describe the response rather than the interference. The bound is invariant under $(v_i, \varepsilon) \mapsto (cv_i, c\varepsilon)$, so $V_{\max} = 1$ simply expresses the budget in units of the largest source weight.

The certificate assumes a meaningful near/far radius, $0 < \varepsilon < A$ for inverting the aggregate envelope, and fixed source amplitudes as σ varies. The single-source reference scale $\sigma_{\mathrm{pair}} = d/\sqrt{2\ln(V_{\max}/\varepsilon)}$ additionally requires $0 < \varepsilon < V_{\max}$ (that is, $0 < \varepsilon < 1$ in the normalised convention). An empty far set is a separate case: its tail is exactly zero, no participation ratio is needed, and the query imposes no bandwidth restriction through this certificate. Any predetermined radius defines a valid partition; we use the tenth percentile of pairwise centre distances as a label-free design choice. Fitted systems whose coefficients change with length scale inherit the bound post hoc at a fixed fit; using it to select their length scale requires an additional coefficient envelope.

Proposition 1 (Certified frontier). *For $A, d > 0$ and $0 < \varepsilon < A$, the largest bandwidth certified by the aggregate envelope is*

$$\sigma^* = \frac{d}{\sqrt{2\ln(A/\varepsilon)}}. \tag{3}$$

For every $\sigma \le \sigma^*$,

$$\mathrm{Tail}_\sigma(y, d) \le A \exp(-d^2/2\sigma^2) \le \varepsilon. \tag{4}$$

The proof uses one path: every far kernel is at most $\exp(-d^2/2\sigma^2)$, and the effective mass converts this per-source envelope into the aggregate bound. Inverting the envelope gives equation (3); monotonicity in σ makes every smaller bandwidth safe. Full derivations and independent numerical checks are in SI Notes 1, 3 and 16.

The mass depends on bandwidth, but it need not be solved jointly with the frontier. Let $\sigma_{\mathrm{pair}} = d/\sqrt{2\ln(1/\varepsilon)}$, the single-source scale, measure A there once, and deploy at equation (3).

Proposition 2 (One-shot conservativeness). *$N_{\mathrm{eff}}(\sigma)$ is nondecreasing in σ. Hence $\sigma^* \le \sigma_{\mathrm{pair}}$, the mass measured at σ_{pair} upper-bounds the deployed mass, and the one-shot frontier satisfies equation* (4).

On the three frozen backbones, the reference measurement exceeds the deployed effective mass by 2.2–4.3× and places the measured tail at 0.08–0.16ε. The exact self-consistent root is only 1.07–1.11× larger and can be found by bisection; the one-shot value is the closed-form certified operating point (SI Note 3).

The Rényi/Hill family D_q [5, 6] sharpens this envelope: $D_2 = N_{\mathrm{eff}}$ is smooth in the weights, whereas D_∞ is the tightest static member, reducing the measured mass by 1.7–2.4× across the three frozen backbones while preserving the budget (SI Note 2). We report D_2 as primary because it is smooth and everywhere-differentiable in the weights (unlike the max in D_∞), which keeps the reported mass and any downstream optimisation well-behaved; the training experiment below regularizes the far mass $m_{\mathcal{F}}$ itself, not D_2.

1.2 Locality controls decisions and deletion

For a classifier, let $S_c^{\mathrm{near}}(y)$ be the score from sources within d and let $m(y)$ be the gap between its two largest class scores. Write the far contribution as a vector $T(y) = \sum_{i\in\mathcal{F}} v_i K_\sigma(\|y - z_i\|^2)$ with class-value vectors $\|v_i\|_1 \le V_{\max}$; the aggregate bound then controls its ℓ_1 norm, $\|T(y)\|_1 \le \varepsilon$ at $\sigma \le \sigma^*$. For a one-hot Parzen field each v_i is a class indicator, so this assumption holds exactly.

Theorem 1 (Decision stability). *Let $\sigma \le \sigma^*$. Because the two-class gap obeys $|T_a - T_b| \le \|T\|_1 \le \varepsilon$, a near-field margin $m(y) > \varepsilon$ makes the deployed and near-field predictions identical.*

Corollary 1 (Performance floor). *Deployed accuracy is at least near-field accuracy minus* $\Pr[m(y) \le \varepsilon]$.

The generic bound with $\|v_i\|_\infty \le V_{\max}$ recovers the earlier 2ε threshold; the sharper ℓ_1 form above is what the one-hot field satisfies, and the main results use it (both norm-dependent forms are stated in SI Note 5).

The same decomposition certifies output sensitivity. A replacement far set is admissible when its sources remain beyond d, its weights obey $\|v_i'\|_1 \le V_{\max}$ and its effective mass does not exceed the calibrated mass.

Corollary 2 (Far-set locality). *At $\sigma \le \sigma^*$, deleting all far sources moves the output by at most ε in ℓ_1. Replacing them by any admissible far set shifts the far contribution by the difference of two tails, of ℓ_1-norm at most 2ε; a near-field margin above 2ε therefore preserves the decision. Arbitrary insertion of additional far sources is not covered: it requires re-calibration or an independent uniform mass bound.*

On four datasets, deleting the far set respects the budget at σ^* and exceeds it by two to three orders of magnitude

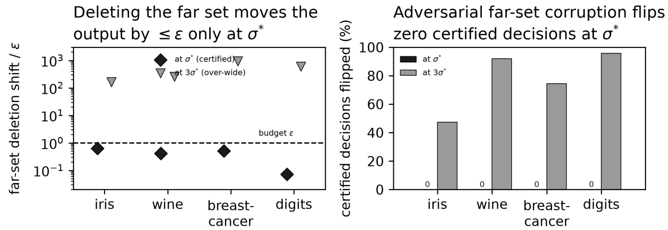

Figure 1: **Certified output locality.** (a) Deleting all far sources moves the output by at most ε at σ^* on all four datasets; $3\sigma^*$ overshoots by two to three orders of magnitude. (b) Under admissible far-set corruption, certified-stable decisions do not flip at σ^*, versus 47–96% at $3\sigma^*$.

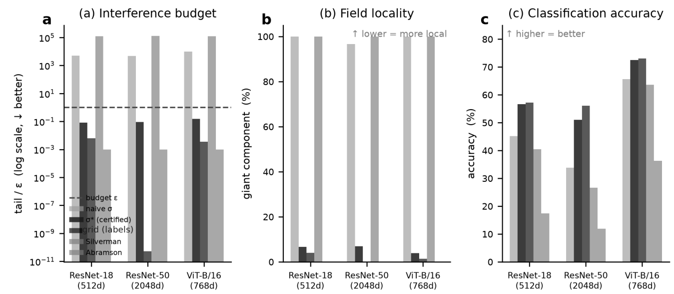

Figure 2: **Certified calibration on three frozen backbones.** (a) Only σ^* holds the interference budget; naive and density bandwidths overshoot by 10^3–$10^5\times$. (b) Those bandwidths collapse the field into one component, whereas σ^* preserves locality. (c) σ^* approaches labelled grid-search accuracy without labels.

at $3\sigma^*$; no certified-stable decision flips under admissible corruption at σ^* (Fig. 1). On a frozen ResNet-18 field, the ℓ_1 margin test certifies 36% of queries and every certified query agrees with its near-field decision. This is a support signal rather than a claim of enhanced accuracy at matched coverage: its distinguishing property is that the selection rule is label-free and fixed by the interference budget.

The guarantee is pointwise when A is measured per query. Our default mean-A construction certifies the mean tail; using an upper quantile certifies the corresponding query fraction and changes accuracy by at most 0.2 points in these experiments. A label-free conformal order statistic gives finite-sample marginal coverage for a future exchangeable query: with 7,000 calibration and 3,000 validation queries at $\alpha = 0.05$, observed coverage is 0.943–0.953 across the three backbones (SI Note 13). This is not a differential-privacy guarantee: it controls the measured field under the stated geometry, not worst-case neighbouring datasets.

1.3 Calibration holds where bandwidth heuristics do not

We build Parzen-vote fields on frozen ResNet-18, ResNet-50 and ViT-B/16 features of CIFAR-100, fixing $\varepsilon = 0.05$ and using no labels to compute σ^* (Methods). The naive single-source bandwidth overshoots the aggregate budget by 4.9×10^3–1.0×10^4 and joins 97–100% of centres into one component. The certified frontier holds tail/ε to 0.08–0.16, reduces the largest component to 4–7%, and reaches accuracies of 56.6%, 51.0% and 72.5%, within 0.6, 5.1 and 0.5 points of the labelled grid-search optima (Fig. 2). Across five re-samples, its accuracy varies by at most 0.3 points.

The median and k-nearest-neighbour heuristics select bandwidths 3.6–6× larger than σ^*, driving tail/ε to approximately 10^5 and losing 9–26 accuracy points. Classical density selectors also miss the budget because they optimise density error rather than interference. These comparisons do not imply the selectors fail at their own objectives; predictive or density fit does not certify aggregate locality. A direct label-free bisection of the measured mean tail finds a bandwidth 1.07–$1.11 \times \sigma^*$, but its 99th-percentile tail exceeds the budget by 1.6–2.3×, so the closed-form frontier is the conservative choice. A hard-truncated Gaussian, which forces exact zero far mass by clipping the kernel at radius d, is the natural same-objective locality baseline; at its labelled-grid bandwidth it reaches 55.8% accuracy (against σ^*'s 56.6%) and agrees with the smooth field on 89% of queries, confirming that exact truncation changes the deployed operator (SI Note 15).

The near sources are 25–47× enriched for the query class relative to chance, although they are not always a majority. Thus the radius identifies a meaningful local vote, but the theorem guarantees which region supplies the decision, not that the decision is correct. Sweeping ε and the radius percentile changes the realised σ^* by only 1.34–1.41× and leaves the certified tail in budget (SI Note 17).

The same pattern reproduces beyond vision. On DBpedia, AG-News, Yahoo-Answers and Covertype, σ^* holds tail/$\varepsilon < 1$ while the naive bandwidth overshoots by 3.6×10^2–2.3×10^3 (Fig. 3). It matches labelled grid search within 0.9 points on the three text sets and trails by 5.8 points on low-dimensional Covertype. On four smaller public datasets it matches the labelled optimum within about two points on three of four. The mechanism is not Gaussian-specific. For a positive decreasing scale kernel $K(r/\sigma)$, one-shot conservativeness holds if and only if $\ln K$ is concave in $\ln r$ (equivalently, the elasticity $xK'(x)/K(x)$ is non-increasing): warming then flattens the normalised far weights in the majorisation order, so every Schur-concave Hill diversity is nondecreasing. Gaussian, Laplace, stretched-exponential, Cauchy and inverse-power tails satisfy the condition (SI Proposition S2). Empirically, the Laplace frontier $\sigma^* = d/\ln(A/\varepsilon)$ holds tail/ε to 0.16–0.24 within 0–2.8 accuracy points of labelled search (SI Note 14).

1.4 Re-calibration follows memory density

Because A is measured rather than fitted, calibration can be repeated when a field changes. We grow each frozen feature field from 3 to 120 examples per class, comparing σ^* re-certified at every stage with a labelled grid-search

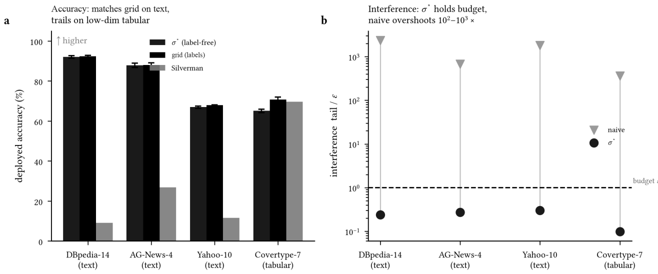

Figure 3: **Cross-modal replication.** (a) σ^* matches labelled grid search on three text datasets and trails on low-dimensional tabular data; error bars are ± 1 s.d. over five seeds. (b) It holds the budget on every corpus, while the naive bandwidth overshoots by two to three orders of magnitude.

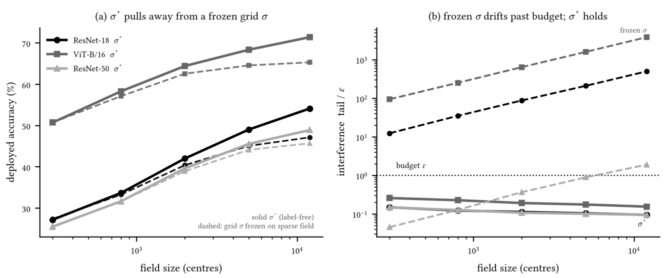

Figure 4: **Re-calibration under densification.** A labelled bandwidth is fixed on the sparse field; σ^* is re-measured without labels. (a) σ^* tracks a per-stage labelled oracle and gains 3.2–7.0 accuracy points over the frozen bandwidth at full density. (b) The frozen bandwidth drifts beyond budget while σ^* remains near 0.1ε.

bandwidth fixed on the initial sparse field. As density increases, σ^* shrinks and holds tail/$\varepsilon \approx 0.1$ (Fig. 4). The frozen bandwidth reaches 502× budget on ResNet-18 and 3,865× on ViT-B/16; re-calibration improves accuracy by 7.0 and 6.1 points, respectively. ResNet-50 is the boundary case: its frozen bandwidth reaches only about 2× budget, while σ^* gains 3.2 points. The radius itself drifts by only 1% over the 40× densification; the dominant change is the measured mass.

This result is adaptivity, not dominance over fresh labelled tuning. Re-certified σ^* tracks a per-stage labelled oracle closely on ResNet-18 and ViT-B/16 and trails it by approximately four points on ResNet-50. It also follows density rather than field size: adding classes at fixed density barely changes the geometry, and the frozen bandwidth then stays in budget and ties σ^*. The null fixes the scope of the growing-memory claim. Recomputing the median and k-nearest-neighbour scales label-free at every density stage, so that all label-free methods receive the updated geometry, does not help: both overshoot the budget by four to five orders of magnitude at every stage, confirming that adapting the scale is not the same as holding the interference budget (SI Note 18).

1.5 Normalised readouts inherit a Gibbs certificate

Additive bounds do not directly control a normalised readout because deleting far sources changes its denominator. Let $p_i = e^{-E_i/\tau}/\sum_j e^{-E_j/\tau}$. Given a partition of the sources into a near set $\mathcal{N}$ and a far set $\mathcal{F}$ – supplied externally, for instance by a task's evidence annotation – let $E_{\mathcal{N}} = \min_{i\in\mathcal{N}} E_i$ be the best near energy and $\Delta = \min_{i\in\mathcal{F}} E_i - E_{\mathcal{N}}$ the energy gap to the nearest far source. If $D_q^{\mathcal{F}}(\tau)$ is the Hill number of the anchor-shifted far weights, the output-relevant far probability mass $m_{\mathcal{F}} = \sum_{i\in\mathcal{F}} p_i$ obeys a direct certificate whenever the gap is positive.

Theorem 2 (Gibbs far-mass certificate)**.** *For any given near/far partition with gap Δ and every $q \geq 2$,*

$$m_{\mathcal{F}}(\tau) \leq D_q^{\mathcal{F}}(\tau) e^{-\Delta/\tau}. \tag{5}$$

If $\Delta > 0$, measure $D_{q,0} = D_q^{\mathcal{F}}(\tau_0)$ once and set

$$\tau_{\mathrm{cert}} = \min\left(\tau_0, \frac{\Delta}{\ln(D_{q,0}/\varepsilon)}\right); \tag{6}$$

then $m_{\mathcal{F}}(\tau_{\mathrm{cert}}) \leq \varepsilon$. If $\Delta \leq 0$ – a far source is at least as favoured as the best near source – the partition is uncertified: no temperature bounds the far mass below ε through this envelope.

For a normalised output $O = \sum_i p_i u_i$ with $\|u_i\| \leq B$, deleting the far set and renormalising changes the output by at most $2B\varepsilon$ (SI Note 11). Unlike the additive form, the Gibbs bound has no source-amplitude factor: it controls the probability mass read by the softmax directly. The theorem therefore applies exactly to a prototypical network [7] with $E_i = \|f(x) - c_i\|^2$ and to attention with $E_i = -\langle q, k_i\rangle$.

On a conv-4 ProtoNet, the reference far mass is 0.080 at $\tau_0 = 1$. For approximately 30% of queries the naive inverted temperature would be warmer than τ_0; the minimum in Theorem 2 is therefore essential. The certified temperature reduces far mass to means of 0.015 for $q = 2$ and 0.017 for $q = \infty$, with maximum $0.0476 < \varepsilon$ over 7,200 queries (Fig. 5a). A separate additive-field experiment confirms the bounded-weight case: the unnormalised prototype tail is 0.25ε at σ^* over 12,000 queries (SI Note 9).

1.6 Task-grounded locality can be diagnosed and trained

Locality is most informative when the near set is fixed externally, not read from the model's own scores. On HotpotQA [8] we use one task-performing reader throughout – `bert-base-uncased` with a pretrained QA adapter, continued-trained on 1,500 retained examples (the matched λ=0 baseline of the training experiment below) – and evaluate on all 1,150 retained held-out examples. The near set is the question and annotated supporting-fact tokens; the far set is the outside-evidence (distractor) tokens, fixed

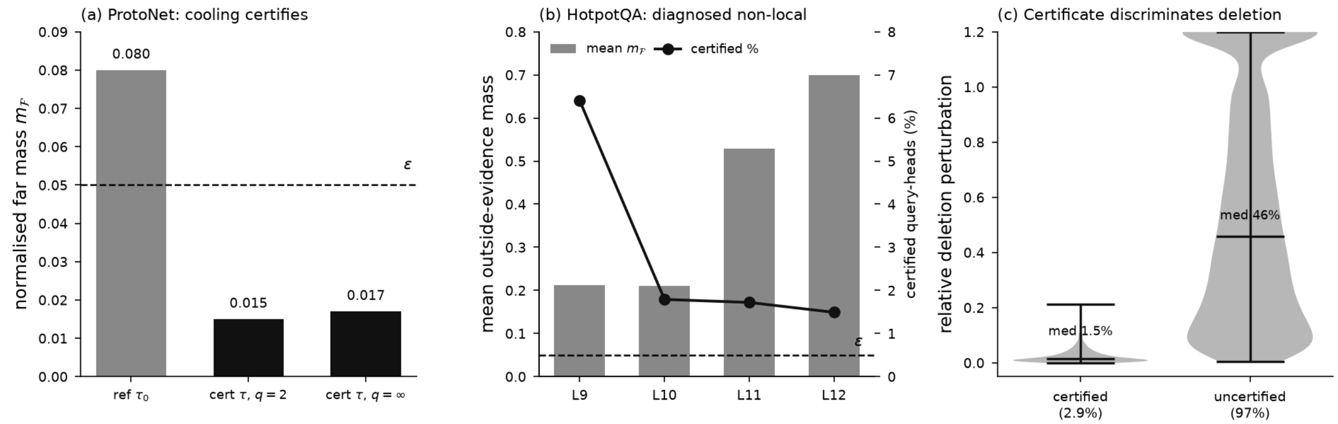

Figure 5: **The operational Gibbs certificate.** (a) A conv-4 prototypical network: cooling from the reference temperature τ_0 to the certified τ_{cert} pulls the normalised far mass from 0.080 below the budget $\varepsilon = 0.05$ (means 0.015 for $q = 2$, 0.017 for $q = \infty$; maximum $0.0476 < \varepsilon$ over 7,200 queries). (b) Task-grounded attention on HotpotQA (fine-tuned BERT reader): with the near set fixed to question and annotated supporting-fact tokens, deployed attention is diagnosed as largely non-local relative to that evidence – mean outside-evidence mass $m_{\mathcal{F}}$ (bars) stays above budget in the upper layers and analytic coverage is 2.9% over all query-heads (4.2% conditional on positive gap; line), with zero bound violations over the positive-gap domain. (c) The certificate is operationally discriminative: deleting the outside-evidence keys moves a certified head's readout by 2.4% on average, against 73% for uncertified heads, so the bound identifies the sub-population of certified-mass heads with bounded, empirically small deletion sensitivity.

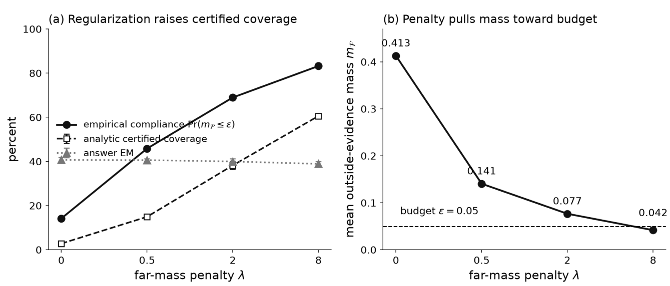

Figure 6: **Locality is trainable (HotpotQA).** Fine-tuning the same reader with a far-mass penalty $\lambda \bar{m}_{\mathcal{F}}$ (backbone frozen, three seeds; bands are seed standard deviations). (a) Analytic certified coverage (unconditional, over all query-heads) rises from 2.8% at the matched λ=0 control to 61%, and empirical compliance $\Pr(m_{\mathcal{F}} \leq \varepsilon)$ from 14% to 83%, while answer exact-match falls only from 0.407 to 0.389. (b) Mean outside-evidence mass falls from 0.41 to 0.042, below the $\varepsilon = 0.05$ budget. Zero bound violations at every λ.

before any attention is read (Fig. 5b,c). The generalized Gibbs bound of Theorem 2 (anchored by the best near-evidence key) holds with zero violations across all positive-gap query-heads (positive-gap rate 0.66), and the deletion bound of SI Note 11 likewise never fails.

The audit separates what the certificate measures from what it guarantees (SI Note 10). Empirical compliance – the fraction of heads whose outside-evidence mass is already at or below $\varepsilon = 0.05$ – is 14%; the stricter analytic coverage, where the closed-form bound itself falls below ε, is 2.9%, and among empirically compliant heads the analytic certificate fires on 20% (its efficiency). Deployed attention is thus largely non-local relative to annotated evidence, with mean outside-evidence mass 0.41 concentrated in the answer-integrating upper layers. Where a head certifies the certificate is operationally *discriminative*: deleting the outside-evidence keys and renormalising moves a certified head's readout by a mean relative 2.4%, against 73% for uncertified heads. This is discrimination, not tightness – the analytic bound overshoots the observed perturbation by a median factor of about six – so the certificate reliably *identifies* the low-interference sub-population rather than sharply predicting each displacement.

The external annotation carries information beyond set size and beyond attention's own concentration. A size-matched random far set certifies far fewer heads (0.5%), and the task partition exceeds it by a paired margin of 2.3 points (example-level bootstrap 95% CI $[2.1, 2.6]$, excluding zero); the endogenous top-decile-logit partition – where "far" is by construction where attention is already smallest, a method sanity check rather than independent evidence – certifies far more (21%). Both controls satisfy the bound with zero violations (SI Note 10). The additive Gaussian representation [9, 10] is in SI Note 8.

Because the far mass is a differentiable function of the attention logits on a fixed annotation partition, it can also be added to the loss. Fine-tuning the same reader (backbone frozen, three seeds) with $L = L_{\text{QA}} + \lambda \bar{m}_{\mathcal{F}}$ raises analytic certified coverage (unconditional, over all query-heads – the same denominator as the 2.9% audit above) from 2.8% at the matched λ=0 control to 61% at λ=8, and empirical compliance from 14% to 83%, pulling mean outside-evidence mass from 0.41 to 0.042, below the $\varepsilon = 0.05$ budget (Fig. 6). The cost in answer accuracy is small: exact-match falls from 0.407 to 0.389 at the strongest penalty (about two seed standard deviations) and is within noise at λ=2 (0.400). This is evidence-supervised far-mass regularization followed by RCP certification, not training the certificate itself; the gold supporting-fact annotation is used during training. Evidence-guided attention supervision is not itself new [33, 34]; the contribution here is the diagnose–regularize–re-certify pipeline and its analytic audit. The diagnosis and the trainability are not specific to this reader or corpus: re-running the whole pipeline over a 2×2 grid of two readers (BERT, RoBERTa) and two evidence-annotated datasets (HotpotQA, 2WikiMultiHopQA), three seeds per cell, reproduces all three findings in every one of the twelve runs—zero bound violations, a task-vs-random certified-coverage margin whose bootstrap CI excludes zero, and a far-mass penalty that raises certified coverage—at an accuracy cost of 0.4–2.0 exact-match points on average (SI Note 10, Table S1).

2 Discussion

The Resolution Calibration Principle changes the role of bandwidth or temperature. Rather than selecting resolution only for predictive fit, it derives the largest operating scale certified to satisfy a locality budget. For additive fields, one measured effective mass gives a closed-form frontier; the same bound controls decision stability, deletion and abstention. Re-measurement adapts the certificate as memory density changes, while the Gibbs form extends it to normalised prototype and attention readouts. The experiments show both outcomes a certificate should expose: ProtoNet can be operated within budget, whereas for a fine-tuned reader on HotpotQA deployed attention is diagnosed as largely non-local relative to annotated task evidence, with a certified sub-population of heads whose outside-evidence mass is certified and whose deletion sensitivity is bounded and empirically small, and where a far-mass penalty trains that locality in. The result certifies locality relative to a fixed evidence annotation; it does not establish a causal explanation of the prediction [11, 12], and the coverage figures are not claimed to transfer across models or tasks.

RCP is a data-dependent certificate: its effective mass is measured from the realised field rather than bounded independently of it, and conditional on that statistic and the stated assumptions the locality implication is mathematical; labels and predictive outcomes enter neither the mass measurement nor the additive calibration. This places it opposite the worst-case certified-robustness paradigm – randomized smoothing and its differential-privacy form [35, 36] certify a fixed classifier against every perturbation in a norm ball under an injected noise distribution, a guarantee that holds regardless of the data; RCP instead certifies a property (far-field interference) of the data as it is, and is silent about perturbations it never measures. The nearest numerical-analysis prior art is error-controlled fast Gaussian summation [30, 31, 32], which holds the bandwidth fixed and selects approximation machinery to control $|\widehat{F} - F|$; RCP instead fixes a declared radius and interference budget and selects the field's operating resolution so that a portion of the exact, untruncated field is below budget. The objective distinguishes this work from bandwidth selectors that minimise density error or predictive loss [1, 2, 13, 14, 4]; such rules can be statistically appropriate while leaving interference uncontrolled. Distance heuristics [15, 16] set a geometric scale but omit the effective-mass safety factor. Direct tail bisection targets the same mean interference empirically, but lacks the one-shot conservative envelope and can leave upper-quantile queries over budget. Conformal and selective prediction [17, 18, 19] address complementary uncertainty questions; here an unlabelled conformal sample calibrates future-query coverage of a geometric mass rather than prediction error. The deletion bound also connects to unlearning and influence attribution [20, 21], but certifies output sensitivity only under the stated bounded-weight and admissibility assumptions. Architectural locality – sparse or windowed attention [22, 23] and hash-bucketed retrieval [24, 25] – imposes a fixed near/far structure by design, whereas the certificate here measures whether an unmodified field already respects an interference budget, and at what resolution. The remaining adjacent tools optimise a different quantity: kernel-based attribution [26] decomposes a prediction per source but does not bound aggregate far influence; kernel-selection [27] targets predictive fit, not a locality budget; retrieval-augmented memories [28] leave the retrieval scale uncalibrated; and data-dependent hashing [29] bounds near-neighbour recall, not far-source influence.

Three limits delimit the claim. The principle requires a meaningful near/far separation and is silent for pure density recovery where every sample is intended to contribute. It calibrates interference, not accuracy; accuracy gains appear in dense growing fields, whereas fixed low-density fields show parity. Finally, one-shot conservativeness assumes fixed source amplitudes and radius as bandwidth varies; representations whose coefficients or geometry co-adapt require re-certification. The construction is radial only relative to the chosen geometry: the same proof applies to any frozen scale-independent dissimilarity, including fixed Mahalanobis and graph-derived distances (SI Note 4). Beyond diagnosis, the differentiable far mass can be regularized during fine-tuning to move a reader's attention into budget (Results, Fig. 6); locality is therefore not only measurable but, on this task, trainable at a small accuracy cost.

3 Methods

Construction. For each backbone—ResNet-18 (512-d), ResNet-50 (2048-d), and ViT-B/16 (768-d), all ImageNet-pretrained and frozen—we extract the penultimate features of CIFAR-100 and build a correction field of 20,000 balanced kernels (200 per class) over the training features, classified by a kernel-weighted (Parzen) vote. The locality radius d is the tenth percentile of pairwise centre distances—pure geometry, no labels. We fix $\varepsilon = 0.05$ a priori and compare the naive single-interferer rule $\sigma_{\text{pair}} = d/\sqrt{2\ln(1/\varepsilon)}$, the certified frontier σ^*, and a labelled grid search over a 30-point log-spaced grid. We also run classical density bandwidths, the median heuristic and a k-nearest-neighbour scale.

Computing σ^*. Four steps require no labels: (1) fix ε and set d to the tenth percentile of pairwise centre distances; (2) form $\sigma_{\text{pair}} = d/\sqrt{2\ln(1/\varepsilon)}$; (3) measure $A = V_{\max}N_{\text{eff}}$ at σ_{pair}, averaging the far-mask participation ratio over queries; and (4) return $\sigma^* = d/\sqrt{2\ln(A/\varepsilon)}$. Proposition 2 makes this measurement conservative. Quantile and conformal variants replace the mean mass by the required upper order statistic. Full grids, seeds and growing-field and modality protocols are in SI Note 18.

Normalised fields. For each query, the Gibbs experiments measure $D_q^{\mathcal{F}}$ at the model's reference temperature τ_0 and deploy or diagnose the pointwise τ_{cert} of Theorem 2 ($q = 2$ primary, $q = \infty$ comparison, $\varepsilon = 0.05$). The ProtoNet uses squared query–prototype distance, with the far gap measured from the nearest prototype. The task-grounded attention experiment uses one HotpotQA distractor-setting reader throughout (`bert-base-uncased` with the pretrained `pf-hotpotqa` adapter, continued-trained on 1,500 retained examples, eager attention), the official development split truncated to 512 tokens, retaining only examples where all annotated supporting sentences and the answer survive (21% retained; 1,150-example held-out test set). The near set is the question, annotated supporting-fact tokens and [CLS], [SEP]; the far set is the remaining distractor context, fixed before attention is read. The analysed layers are the upper four, all heads reported. The generalized bound anchors $D_q^{\mathcal{F}}$ at the best near-evidence key (SI Note 10); the training experiment regularizes the far mass with backbone frozen over three seeds. Architectural, audit, control and numerical details are in SI Notes 8–12.

Hard-truncation comparator. As a same-objective locality baseline we run a hard-truncated Gaussian $K_\sigma^{\text{trunc}}(r^2) = K_\sigma(r^2)\mathbf{1}\{r \le d\}$, whose bandwidth is chosen by the same labelled oracle grid. It has exact zero far mass by construction, so it changes the readout rather than calibrating it. We compare accuracy and decision agreement with the smooth field; full results in SI Note 15.

Frozen-geometry stress test. To probe the frozen-geometry corollary (SI Note 4) we build controlled anisotropic-Gaussian, two-moons and densification settings and run RCP under isotropic-Euclidean, frozen global-Mahalanobis and frozen out-of-sample graph-geodesic dissimilarities, holding each geometry fixed during calibration. We report tail budget, per-query violation rate and decision accuracy in SI Note 17.

Reproducibility. The full computational specification, proofs, supplementary displays and released outputs are in the Supplementary Information and repository. Proofs of the additive frontier, one-shot monotonicity, decision and deletion corollaries, Rényi family, Gibbs certificate and conformal query certificate are given in SI Notes 1–18.

Data and code availability. All experiment scripts and released result files are available at `github.com/negulescu42/sigma-star`, with fixed seeds and the exact bandwidth grid for each experiment.

Author contributions. R.N. conceived the resolution-calibration framing, developed the empirical and machine-learning programme, and carried out the transformer, prototype-network and re-calibration experiments. M.T. performed the mathematical audit and contributed the vector ℓ_1 decision certificate and the sharpened statements of the additive results. C.B. developed the analytic core, including Propositions 1–2, Theorem 1, Corollary 2, and the Rényi, Gibbs and conformal extensions. All authors reviewed and approved the manuscript.

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